

Heimdyn

Pallet Manufacturing ERP

Scope of Work

Phase 2 — Building the connected manufacturing engine

Project	Heimdyn Factory ERP — Manufacturing Module
Prepared For	Advance Lumber & Pallet Ltd. · Surrey, BC
Phase	Phase 2 — ERP Core (Showcase Build)
Prepared By	Heimdyn Development Team
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1 Executive Summary

This document sets out exactly what we will build in Phase 2 of the Heimdyn platform: a streamlined Enterprise Resource Planning (ERP) system purpose-built for a wooden pallet manufacturer.

Phase 1 delivered a factory operations dashboard. Phase 2 brings the business to life — connecting sales, manufacturing, purchasing and inventory into one flowing system where a single customer enquiry can ripple through to a production order, a material purchase, and a finished delivery, all automatically linked.

The design philosophy is deliberate: take the proven logic of a full MRP system, and strip it down to a clean, intuitive core that an operations team can pick up in minutes — not weeks. Every screen prioritises clarity over clutter.

Phase 2 — What we build now

A fully working ERP interface covering the complete order-to-delivery and purchase-to-stock journeys, ready for live demonstration. Built on realistic sample data.

Phase 3 — What comes next

Connecting the system to a live, secure database with multi-location support, and full onboarding of Advance Lumber & Pallet as a live client.

At a glance

5 Modules

Quotations, Sales, Manufacturing, Purchasing & Inventory — all talking to each other.

6 User Roles

Clear responsibilities and approval gates for every person in the workflow.

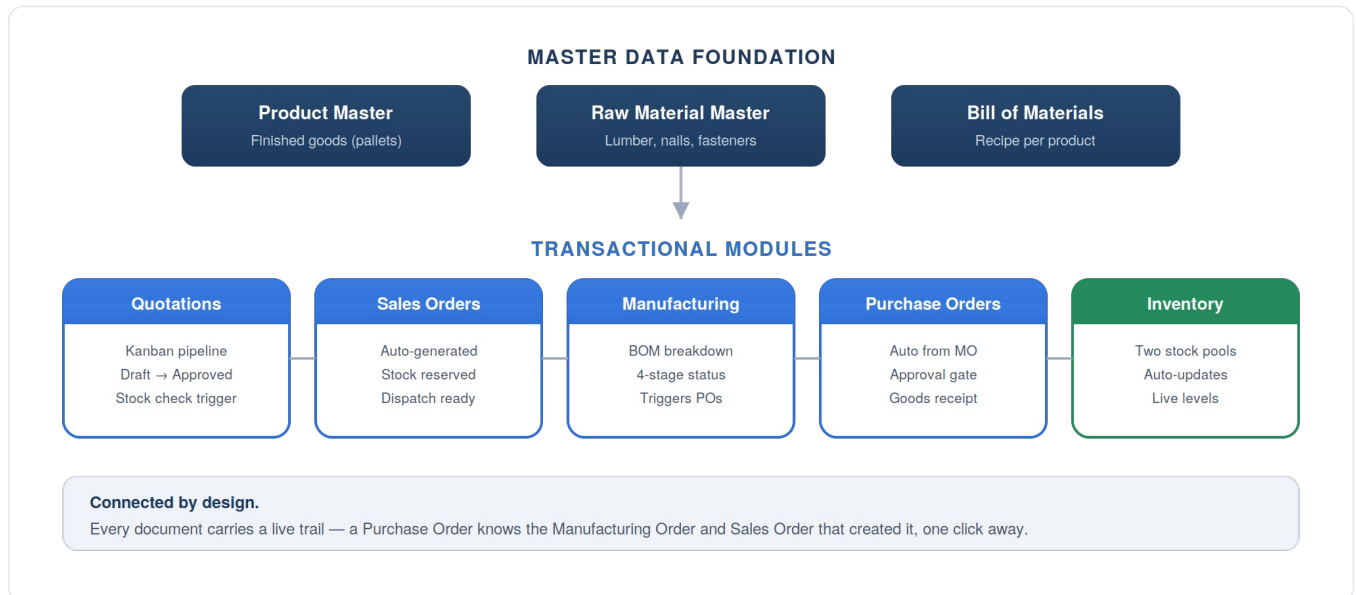
2 Pallet Lines

New and recycled pallets, the two core products in scope.

2 How the System Works

Everything starts with three master records and flows through five connected modules.

Think of the master data as the foundation — the catalogue of what you sell, what you build with, and the recipe that links the two. On top of that sit the five working modules where the day-to-day business happens.



The Heimdyn ERP at a glance — foundation data feeding five connected modules.

The single most important idea in this system is **connection**. In most basic software, a sales record and a purchase record live in separate worlds. Here, they are linked end to end. Open any document and you can trace its entire family tree — the sale that started it, the production order it triggered, and the material purchase that fed it.

3 The Core Business Flow

There are two routes a customer order can take. The system decides automatically which one applies, based on whether the product is already in stock.

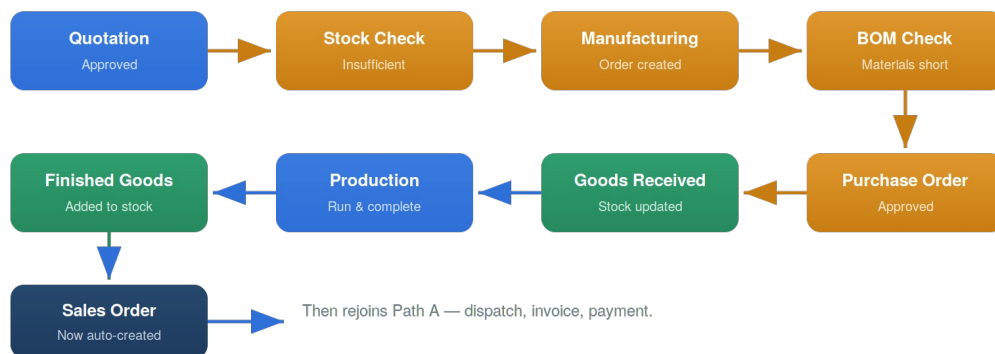
Path A — Stock Available

The straightforward route: quotation to cash, no manufacturing required.



Path B — Manufacturing Required

When stock runs short, the system spins up production and procurement automatically.



Two paths, one system — stock-available orders ship straight away; short-stock orders trigger production and purchasing.

3.1 Path A — Stock is available

The simplest journey. The customer wants something we already have on the shelf.

- 1 A Sales Executive creates a **quotation**, choosing the pallet and quantity. The system checks stock in real time as they type.
- 2 The quotation is submitted, and the **Sales Manager approves** it.
- 3 The system confirms **stock is sufficient** and automatically raises a **Sales Order**, marking that stock as committed to the order.
- 4 Goods are **dispatched**, and inventory is reduced automatically.
- 5 **Accounts** raises the tax invoice and records payment. Done.

3.2 Path B — Manufacturing is required

When stock runs short, the system quietly does the heavy lifting — spinning up production and ordering materials without anyone chasing paperwork.

- 1 The quotation is approved as before, but the **stock check comes back short**.
- 2 The system automatically creates a **Manufacturing Order** and sends it to the Production Manager.
- 3 On approval, the system checks the **recipe (BOM)** against raw material stock. Where materials are short, it **auto-raises Purchase Orders**.

- 4 The **Purchase Manager approves** the orders; materials arrive and are **received into stock**.
- 5 **Production runs**, the order is marked complete, and finished pallets flow into inventory.
- 6 The **Sales Order is now created automatically** — and the order rejoins Path A for dispatch and invoicing.

The golden rule

A Sales Order is never created by hand. The system always creates it — either immediately when stock exists, or after manufacturing finishes when it doesn't. This keeps the numbers honest and removes guesswork.

4 Master Data Foundation

Three foundational records hold everything together. These are set up once, at the start.

Product Master — what you sell

Every sellable, buildable pallet. Only physical products are in scope — new pallets and recycled pallets.

Detail captured	Example
Product name & code	Euro Pallet Standard 1200×800mm · EPAL-100
Type	New Pallet / Recycled Pallet
Price (USD)	Selling price in USD
Status	Active / Inactive

Raw Material Master — what you build with

Every material consumed in production — lumber, nails, fasteners, treatment labels.

Detail captured	Example
Material name & code	Pine Planks 2×4, Steel Nails 2"
Unit of measure	metres, units, kilograms, gallons
Current stock	Opening quantity on hand

Bill of Materials — the recipe

For each product, a fixed recipe listing the materials and quantities needed to build one unit. In Phase 2 these use realistic sample data for pallet construction.

5 The Modules in Detail

5.1 Quotations

The starting point of every sale. Presented as a visual pipeline board, so the whole team can see at a glance where every enquiry stands.

Draft → **Pending Approval** → **Proforma Invoice** → **Sales Order Raised**

- Sales Executives build quotations; the Sales Manager approves them.
- A live stock indicator shows green when items are in stock, amber when production will be needed.
- Cards needing manufacturing are flagged **Requires MO** right on the board.
- Pricing shown in USD on every quote.

5.2 Sales Orders

Confirmed business, ready to fulfil. Always created by the system — never by hand.

Confirmed → **Stock Committed** → **Dispatched** → **Invoiced**

Each Sales Order carries its full history — the quotation it came from, and the manufacturing order behind it where relevant.

5.3 Manufacturing Orders

Where production is planned and tracked. Created automatically when stock is short, or manually by the Production Manager to build ahead.

Pending Approval → **Planned** → **In Progress** → **Done**

- A built-in BOM breakdown shows exactly which materials are sufficient and which fall short.
- Short materials are flagged instantly, and the system shows a **PO Raised** badge once it has ordered them.
- A progress indicator shows units completed against the target, updated by the Production Manager.

Inventory moves automatically

- Raw materials are consumed the moment an order moves to **In Progress**.
- Finished pallets are added to stock the moment an order is marked **Done**.

5.4 Purchase Orders

Material procurement, triggered automatically by production shortfalls. The Purchase Manager approves and receives.

Pending Approval → **Approved** → **Goods Received**

- Generated automatically when a manufacturing order is short on materials — no manual creation needed.
- A summary bar displays open orders and total pending value.
- Recording goods receipt updates raw material stock instantly.

5.5 Inventory

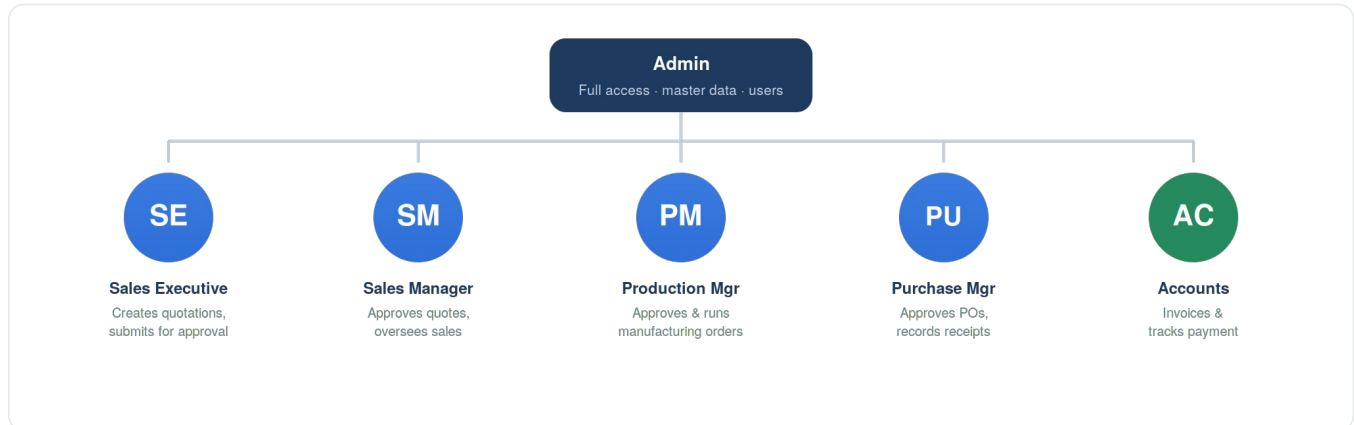
Two clearly separated stock pools that keep themselves up to date.

Stock Pool	Goes down when	Goes up when
Finished Goods	An order is dispatched	Manufacturing completes
Raw Materials	Production begins (consuming the recipe)	Goods are received, or stock is added manually

The only manual action is topping up raw materials directly — useful for recycled pallet inputs that arrive outside the purchasing system.

6 Who Does What — User Roles

Six roles, each with clear responsibilities and the right approval gates. People see only what they need.



Clear lines of responsibility, with Admin overseeing the whole system.

Role	Responsibilities
Admin	Full access. Manages all master data and users. Can act across every module.
Sales Executive	Creates and submits quotations. Views order status and stock availability. Cannot approve.
Sales Manager	Approves or rejects quotations. Oversees all sales activity and analytics.
Production Manager	Approves manufacturing orders and drives them through production to completion.
Purchase Manager	Approves purchase orders and records incoming goods against them.
Accounts	Raises tax invoices from confirmed orders and tracks payment. View-only on operations.

7 What's Included — Deliverables

The complete set of working components we will hand over at the end of Phase 2.

#	Deliverable	How we'll know it's done
1	Product Master with full management	Products can be added, edited, deactivated
2	Raw Material Master with full management	Materials can be added, edited, deactivated
3	Bill of Materials editor	Recipes can be defined and updated per product
4	Quotations module (board, detail view, 3-step creation)	Pipeline visible, creation works, stock check live
5	Sales Orders module with auto-creation	Orders appear automatically with correct status flow
6	Manufacturing Orders with approvals & BOM view	Auto-created, approvable, material check visible
7	Purchase Orders with approvals & goods receipt	Auto-raised, approvable, receipt updates stock
8	Inventory with two stock pools	Accurate levels, manual top-up works
9	Dashboard updated with live ERP figures	KPIs, pipeline bars and activity feed all working
10	Role-based access for all six roles	Each role sees only its permitted actions
11	Document Trail across every module	Linked history visible in each detail view
12	USD currency support	Pricing and totals shown in USD across all documents
13	Sample data across the whole system	Demo-ready out of the box

8 What's Not Included

To keep Phase 2 focused and deliverable, the following are intentionally left out. Each can be added later through a simple change request.

Area	Not included in Phase 2
Other product lines	Pallet repairs (a service) and pre-cut lumber (trading). Only new and recycled pallets are in scope.
Multiple locations	The system assumes a single factory and warehouse for now.
Live database	Phase 2 runs on realistic sample data; the live database arrives in Phase 3.
Automatic reordering	No automatic stock-level reorder rules. Purchases are triggered only by production needs.
Accounting integration	No link to accounting software, general ledger or payment gateways.
Customer returns	No inbound returns flow; recycled pallet inputs are entered manually.
Custom recipes	Bills of materials are fixed per product; custom-dimension recipes come later.
Existing analytics pages	The original dashboard analytics pages stay as they are.

9 Assumptions & Dependencies

9.1 Assumptions

- All data in Phase 2 is realistic sample data, to be replaced with live data in Phase 3.
- Bill of materials definitions will use realistic pallet manufacturing data provided by our team.
- The approved Heimdyn design system and prototype screens are the reference for all visuals.
- No external integrations are required at this stage.
- Vendor records for purchasing use the existing vendor module from Phase 1.
- A Sales Order is created automatically by the system — never by hand. Once a quotation is approved (and stock is ready), the system raises the Sales Order itself. Staff cannot manually create or "convert" one.

9.2 Dependencies

- The Phase 1 platform must be stable and deployed before Phase 2 begins.
- The existing users module will be extended to include the six new roles.
- The design system remains fixed; any visual changes need a separate review.

10 Managing Change

Anything outside this document follows a simple, transparent process — so there are never surprises on either side.

- 1 A written change request describes what's wanted and why.
- 2 We review the impact on timeline and approach.
- 3 If agreed, this document is updated, versioned and re-signed.
- 4 No out-of-scope work begins without that sign-off.

11 Agreement & Sign-off

By signing below, both parties agree to the scope described in this document. Development of Phase 2 begins once signatures are in place.

Prepared By — Heimdyn Team

Name _____

Signature _____

Date _____

Approved By — Project Lead

Name _____

Signature _____

Date _____

Version	Date	Author	Summary
1.0	May 2026	Heimdyn Development Team	Initial scope — Phase 2 ERP defined